Baseline (Logistic Regression)

In Model 1, Logistic Regression is used as a baseline method. This model utilizes only the original input features without considering any interaction or correlation between them. Each feature contributes independently to the prediction. This approach represents traditional factor-based methods commonly used in landslide risk prediction. The performance of this model serves as a reference point to evaluate the effectiveness of more advanced approaches.

```python
import pandas as pd  # veri işlemleri için kütüphane

from sklearn.model_selection import train_test_split  # veriyi train/test ayırmak için
from sklearn.linear_model import LogisticRegression   # Logistic Regression modeli
from sklearn.metrics import accuracy_score, classification_report  # performans ölçümleri

# Feature (girdi) ve label (çıktı) ayır
X = df.iloc[:, :-1]   # son sütun hariç tüm sütunlar → feature
y = df.iloc[:, -1]    # son sütun → hedef (0 veya 1)

# Veriyi eğitim ve test olarak böl (%70 train, %30 test)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Model oluştur
model = LogisticRegression(max_iter=1000)
# max_iter → modelin daha iyi öğrenmesi için iterasyon sayısı

# Modeli eğit
model.fit(X_train, y_train)

# Test verisi üzerinde tahmin yap
y_pred = model.predict(X_test)

# Model performansını yazdır
print("Accuracy:", accuracy_score(y_test, y_pred))
# Accuracy → doğru tahmin oranı

print("\nClassification Report:\n", classification_report(y_test, y_pred))
# Precision, Recall, F1-score gibi detaylı metrikler
```

Result :

```
Accuracy: 0.7333333333333333

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.81      0.82       441
           1       0.50      0.51      0.50       159

    accuracy                           0.73       600
   macro avg       0.66      0.66      0.66       600
weighted avg       0.74      0.73      0.73       600
```

improved model (GBDT + Feature Interaction)

In Model 2, a Gradient Boosting Decision Tree (GBDT) model is applied along with feature interaction techniques to capture correlations between input variables. Polynomial feature expansion is used to generate interaction terms between features, allowing the model to learn complex relationships. This approach simulates the correlation-aware mechanism described in the knowledge graph-based method. As a result, the model is expected to achieve higher prediction performance compared to the baseline model.

```python
import pandas as pd  # veri işlemleri

from sklearn.model_selection import train_test_split  # veri bölme
from sklearn.ensemble import GradientBoostingClassifier  # GBDT modeli
from sklearn.metrics import accuracy_score, classification_report  # performans ölçümleri
from sklearn.preprocessing import PolynomialFeatures  # feature interaction oluşturmak için

# Feature ve label ayır
X = df.iloc[:, :-1]   # input veriler
y = df.iloc[:, -1]    # hedef (0 veya 1)

# 🔥 Feature interaction oluştur (korelasyon simülasyonu)
poly = PolynomialFeatures(
    degree=2,               # 2'li kombinasyonlar (çarpımlar)
    interaction_only=True,  # sadece feature'lar arası çarpım (x1*x2)
    include_bias=False      # sabit 1 ekleme
)

X_interaction = poly.fit_transform(X)
# Yeni feature seti:
# örnek → x1, x2, x1*x2, x1*x3, ...

# Train - Test ayır
X_train, X_test, y_train, y_test = train_test_split(
    X_interaction, y, test_size=0.3, random_state=42
)

# GBDT modeli oluştur
model = GradientBoostingClassifier(
    n_estimators=100,   # ağaç sayısı
    learning_rate=0.1   # öğrenme hızı
)
```

```python
# Modeli eğit
model.fit(X_train, y_train)

# Tahmin yap
y_pred = model.predict(X_test)

# Sonuçları yazdır
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Result :

```
Accuracy: 0.7516666666666667

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.84      0.83       441
           1       0.53      0.50      0.51       159

    accuracy                           0.75       600
   macro avg       0.68      0.67      0.67       600
weighted avg       0.75      0.75      0.75       600
```

Comparison of Baseline and Improved Model

The experimental results demonstrate that the improved model outperforms the baseline method across most evaluation metrics. The overall accuracy increased from 73.3% in the baseline method to 75.2% in the improved model, indicating a clear enhancement in prediction performance.

In terms of precision, the improved model shows better reliability compared to the baseline method, meaning that its predictions are more accurate when identifying positive cases. Similarly, the F1-score, which reflects the balance between precision and recall, is also higher in the improved model, indicating a more stable and effective performance.

Although there is a slight variation in recall, the overall performance improvements suggest that the improved model is more capable of capturing meaningful patterns in the data. This improvement is mainly due to the incorporation of feature interactions, which allow the model to learn relationships between input variables rather than treating them independently.

Therefore, the results clearly demonstrate that using feature interaction (correlation) significantly improves the performance of the model compared to the baseline method. This confirms the importance of considering relationships between features in landslide risk prediction tasks.

Metric	Baseline Model	Improved Model	Fark
Accuracy	0.733	0.752	+0.019
Baseline Recall	0.81	0.84	+0.03
Improved Precision	0.50	0.53	+0.03
Weighted Avg Recall	0.73	0.75	+0.02